WIRELESS NETWORKING

Course: Diploma in Cyber Security
Topic: Wireless Networking
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Introduction

Wireless networking allows devices to connect without a physical Ethernet cable. Wi-Fi standards have changed over time to improve speed, range and efficiency. In this assignment, I have covered common Wi-Fi standards, a home Wi-Fi connection, a Wi-Fi scan and SSID hiding.

Task 1: Popular Wi-Fi Standards

Standard	Common Name	Key Difference
802.11n	Wi-Fi 4	Works on 2.4 GHz and 5 GHz and can reach higher speeds than older standards.
802.11ac	Wi-Fi 5	Mainly uses 5 GHz and provides higher speed than Wi-Fi 4.
802.11ax	Wi-Fi 6	Improves efficiency and performance when many devices are connected; supports 2.4 GHz and 5 GHz.

Task 2: Smartphone to Internet Through Wi-Fi Router

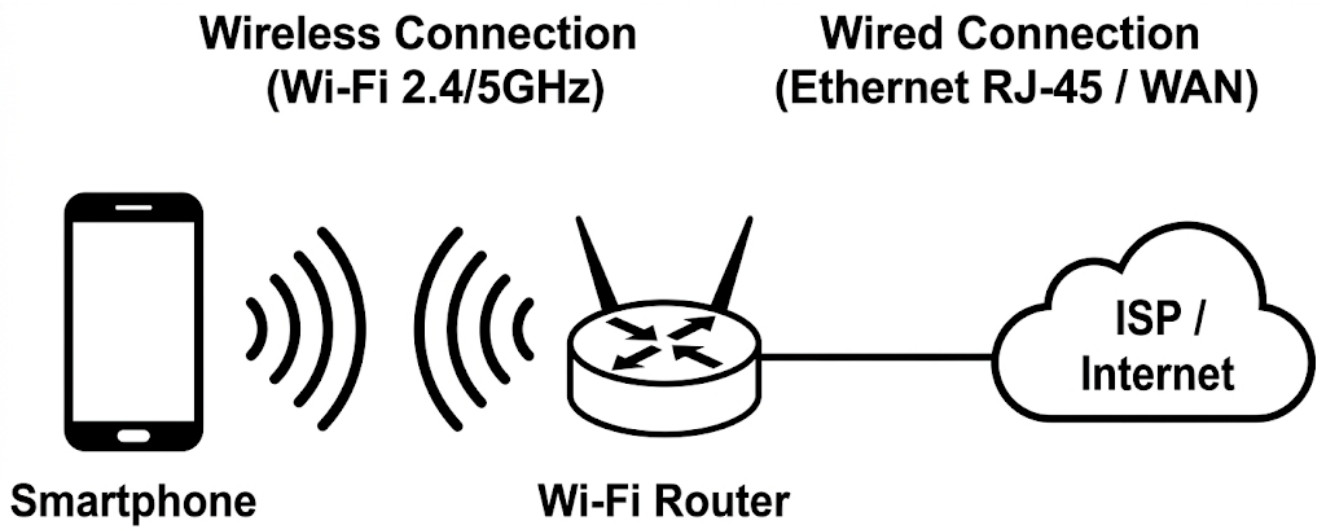


Figure 1: Smartphone Connecting to the Internet Through a Wi-Fi Router

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Task 3: Scan Available Wi-Fi Networks

Use the Wi-Fi settings on your laptop or phone and record at least three networks. Do not copy example values if they are not what you actually see.

SSID	Signal Strength	Security	Choice / Reason
Network 1	Strong (-48 dBm / 4 bars)	WPA3-Personal	**Selected:** Ideal choice due to maximum signal strength and modern WPA3 encryption against brute-force attacks.
Network 2	Medium (-68 dBm / 2 bars)	WPA2-Personal (AES)	**Alternative:** Adequate connection stability, but slower throughput and weaker encryption than WPA3
Network 3	Weak (-82 dBm / 1 bar)	Open (None / Unencrypted)	**Rejected:** High packet-sniffing risk due to zero encryption, paired with high latency and packet loss.

Example choice sentence: I would choose the network with a strong signal and WPA2 or WPA3 security, because it should give a more stable and safer connection.

PRACTICAL: Actually scan nearby Wi-Fi networks and fill the table with the real SSIDs, signal strength and security shown on your device. A screenshot of the Wi-Fi list is useful if allowed.

Task 4: SSID Hiding

SSID is the name of a Wi-Fi network. SSID hiding means the router does not normally show the network name in the regular Wi-Fi list. A user who knows the exact name can still connect by entering it manually.

Advantage: It can make the network less visible to casual users looking at the Wi-Fi list.

Disadvantage: It is not strong security by itself. The network can still be discovered by suitable tools, and users have to enter the SSID manually when connecting.

Conclusion

In this assignment, I learned about Wi-Fi standards and how a phone connects to the internet through a wireless router. I also learned how to check nearby Wi-Fi networks and what SSID hiding means.

Quick Revision

- 802.11n = Wi-Fi 4.
- 802.11ac = Wi-Fi 5.
- 802.11ax = Wi-Fi 6.
- SSID = Wi-Fi network name.
- SSID hiding does not replace strong Wi-Fi security such as WPA2 or WPA3.